

Paul Graf

in collaboration with

Lexin Ding

Julia Liebert

Florian Matz

Markus Penz

Christian Schilling

On an Attempt to Capture Strong Electron Correlation at Mean-Field Computational Cost

Determining the ground state

$$\hat{H}_{\text{elec}} = \underbrace{\left(-\frac{\hbar^2}{2m_e} \sum_i^{\text{elec}} \nabla_i^2 \right)}_{\text{kinetic energy}} + \underbrace{\left(-\frac{e^2}{4\pi\epsilon_0} \sum_i^{\text{elec}} \sum_j^{\text{nuc}} \frac{Z_j}{|\mathbf{r}_i - \mathbf{R}_j|} \right)}_{\text{nuclear potential energy}} + \underbrace{\left(\frac{e^2}{4\pi\epsilon_0} \sum_{i < i'}^{\text{elec}} \frac{1}{|\mathbf{r}_i - \mathbf{r}_{i'}|} \right)}_{\text{Coulomb interaction energy}}$$

The N electron correlation problem.

$$E_0 = \min_{|\Psi_N\rangle \in \mathcal{H}_N} \langle \Psi_N | \hat{H} | \Psi_N \rangle$$



N particle Hilbert
space **scales**
exponentially!



Approximation: Hartree-Fock method implements **mean-field** ansatz as **variation over Slater determinants**

$$|\Psi_N\rangle \in \mathcal{H}_N \quad \longrightarrow \quad |i_1, \dots, i_N\rangle \in \mathcal{H}_N$$

Slater determinants

$$E_{\text{HF}} = \min_{|i_1, \dots, i_N\rangle} \left\{ \sum_i \langle i | \hat{h} | i \rangle + \frac{1}{2} \sum_{i,j} \left[\langle i | \otimes \langle j | \hat{W} | i \rangle \otimes | j \rangle - \langle i | \otimes \langle j | \hat{W} | j \rangle \otimes | i \rangle \right] \right\}$$

Hartree

Exchange

Approximation: Hartree-Fock method implements **mean-field** ansatz as **variation over Slater determinants**

$$|\Psi_N\rangle \in \mathcal{H}_N \quad \longrightarrow \quad |i_1, \dots, i_N\rangle \in \mathcal{H}_N$$

$$E_{\text{HF}} = \min_{|i_1, \dots, i_N\rangle} \left\{ \sum_i \langle i | \hat{h} | i \rangle + \frac{1}{2} \sum_{i,j} \left[\langle i | \otimes \langle j | \hat{W} | i \rangle \otimes | j \rangle - \langle i | \otimes \langle j | \hat{W} | j \rangle \otimes | i \rangle \right] \right\}$$

Functional of the **1-particle reduced density matrix**

$$E_{\text{HF}} = \min_{\rho_1} \left\{ \text{Tr}_1 [\hat{h} \rho_1] + \text{Tr}_2 \left[\hat{W} \left(\frac{1 - \hat{X}}{2} \rho_1 \otimes \rho_1 \right) \right] \right\}$$

$$\rho_1 = N \text{Tr}_{N-1} [\rho_N]$$

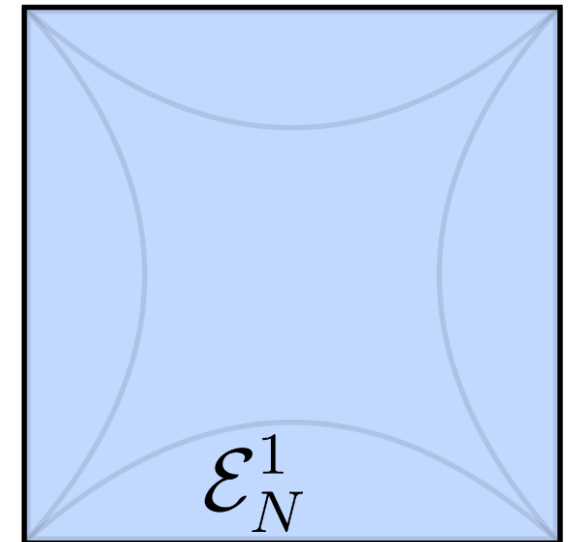
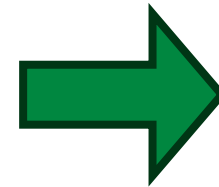
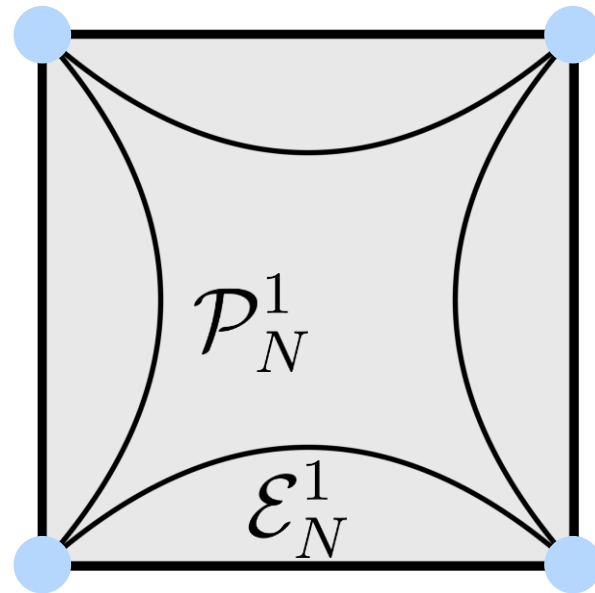
$$\rho_1 = \sum_i n_i |i\rangle \langle i|$$

Approximation: Hartree-Fock method implements **mean-field** ansatz as **variation over Slater determinants**

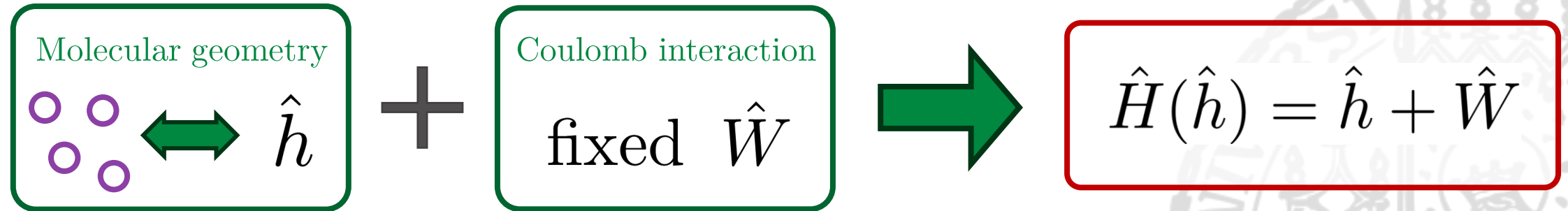
Functional of the **1-particle reduced density matrix**

$$E_{\text{HF}} = \min_{\rho_1} \left\{ \text{Tr}_1 [\hat{h} \rho_1] + \text{Tr}_2 \left[\hat{W} \left(\frac{1 - \hat{X}}{2} \rho_1 \otimes \rho_1 \right) \right] \right\}$$

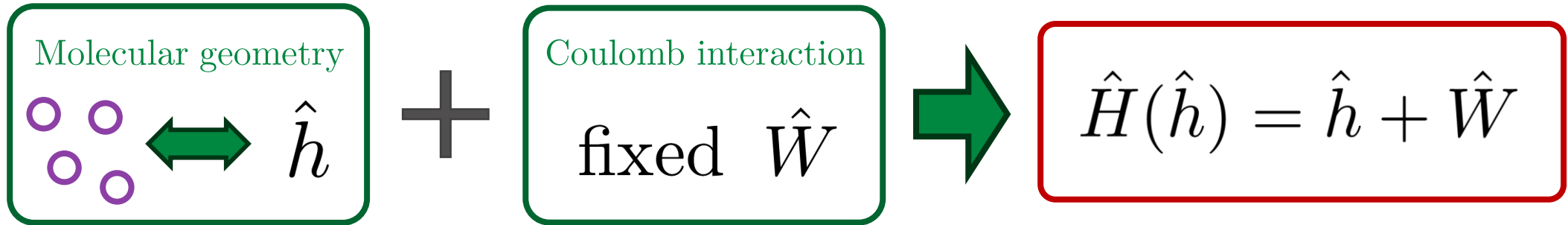
Lieb[1]: Variational domain



1-Particle Reduced Density Matrix Functional Theory *1RDMFT*



1-Particle Reduced Density Matrix Functional Theory *1RDMFT*



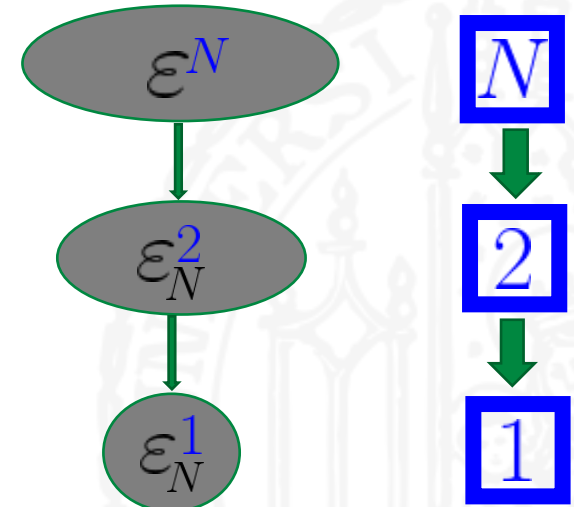
$$E_0(\hat{h}) = \min_{\rho_N} \text{Tr}_N [\hat{H} \rho_N]$$

$$= \min_{\rho_2} \text{Tr}_2 [\hat{H} \rho_2]$$

$$= \min_{\rho_1} \left\{ \text{Tr}_1 [\hat{h} \rho_1] + \min_{\rho_2 \mapsto \rho_1} \text{Tr}_2 [\hat{W} \rho_2] \right\}$$

$$\equiv \min_{\rho_1} \left\{ \text{Tr}_1 [\hat{h} \rho_1] + \mathcal{F}_{\hat{W}}(\rho_1) \right\}$$

fixed



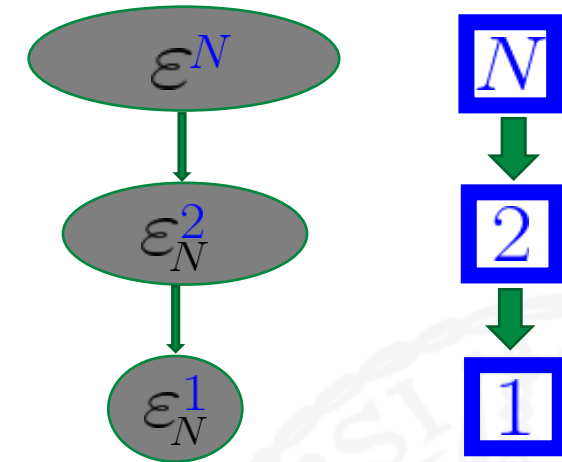
$\mathcal{F}(\rho_1)$ is universal

1-Particle Reduced Density Matrix Functional Theory *1RDMFT* [2-4]

$$\begin{aligned}
 E_0(\hat{h}) &= \min_{\rho_N} \text{Tr}_N [\hat{H} \rho_N] \\
 &= \min_{\rho_2} \text{Tr}_2 [\hat{H} \rho_2] \\
 &= \min_{\rho_1} \left\{ \text{Tr}_1 [\hat{h} \rho_1] + \min_{\rho_2 \mapsto \rho_1} \text{Tr}_2 [\hat{W} \rho_2] \right\} \\
 &\equiv \min_{\rho_1} \left\{ \text{Tr}_1 [\hat{h} \rho_1] + \mathcal{F}_{\hat{W}}(\rho_1) \right\}
 \end{aligned}$$



$\mathcal{F}(\rho_1)$ is universal



Variational
principle

+

Structure / Scope
 $\hat{H}(\hat{h}) = \hat{h} + \hat{W}$



Functional
theory

[2] – M. Levy, *Proc. Natl. Acad. Sci. U.S.A.* **76**, 6062 (1979).

[3] – E. H. Lieb, *J. Quantum Chem.* **24**, 243 (1982).

[4] – J. Liebert, A. Y. Chaou, C. Schilling,
J. Chem. Phys. **158**, 214108 (2023).

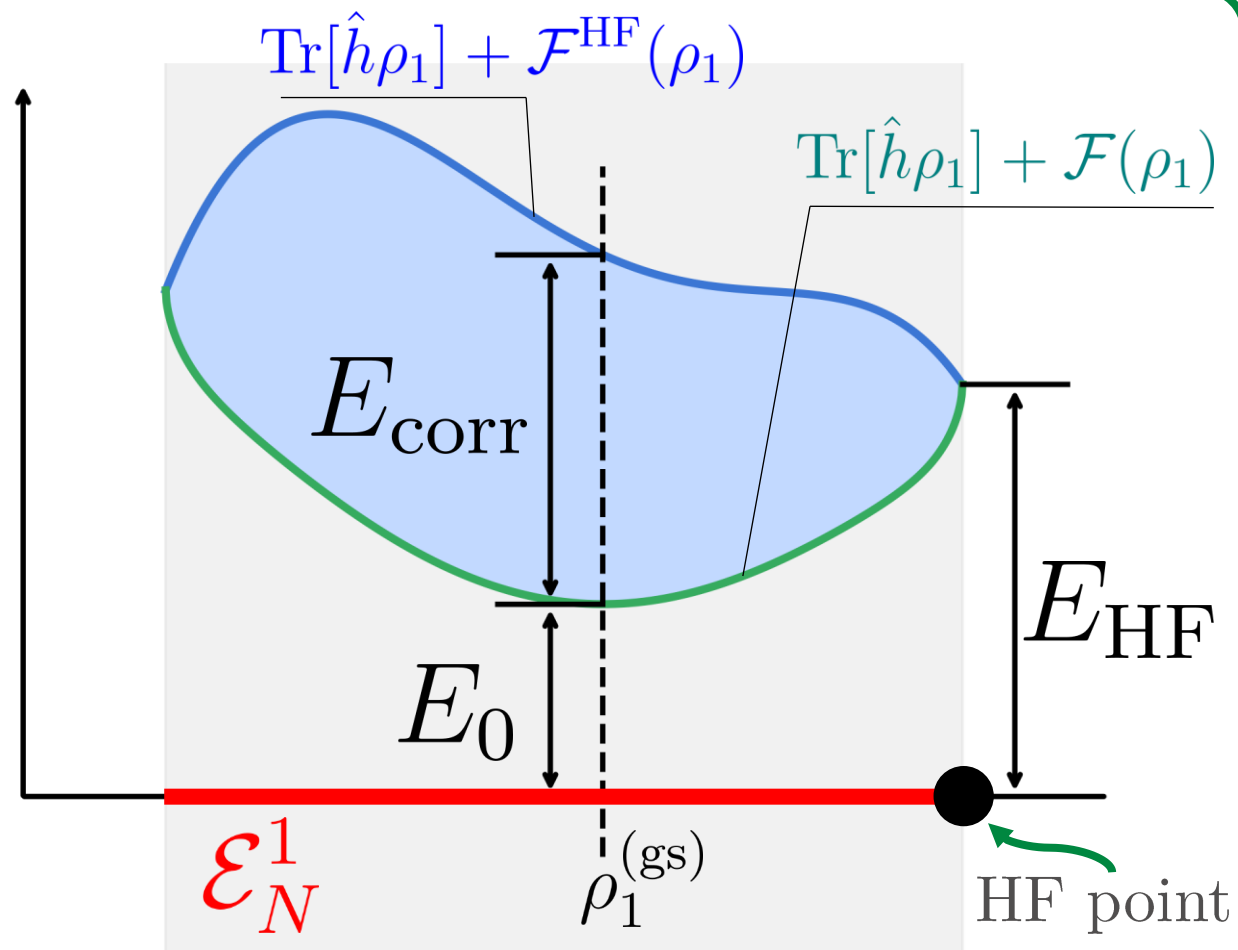
Central challenge of 1RDMFT: obtaining universal functional $\mathcal{F}(\rho_1)$

Partitioning the functional

$$\mathcal{F}(\rho_1) = \mathcal{F}^{\text{HF}}(\rho_1) + \mathcal{F}^{\text{c}}(\rho_1)$$

with

$$\mathcal{F}^{\text{HF}}(\rho_1) = \text{Tr}_2 \left[\hat{W} \frac{1-X}{2} \rho_1 \otimes \rho_1 \right]$$



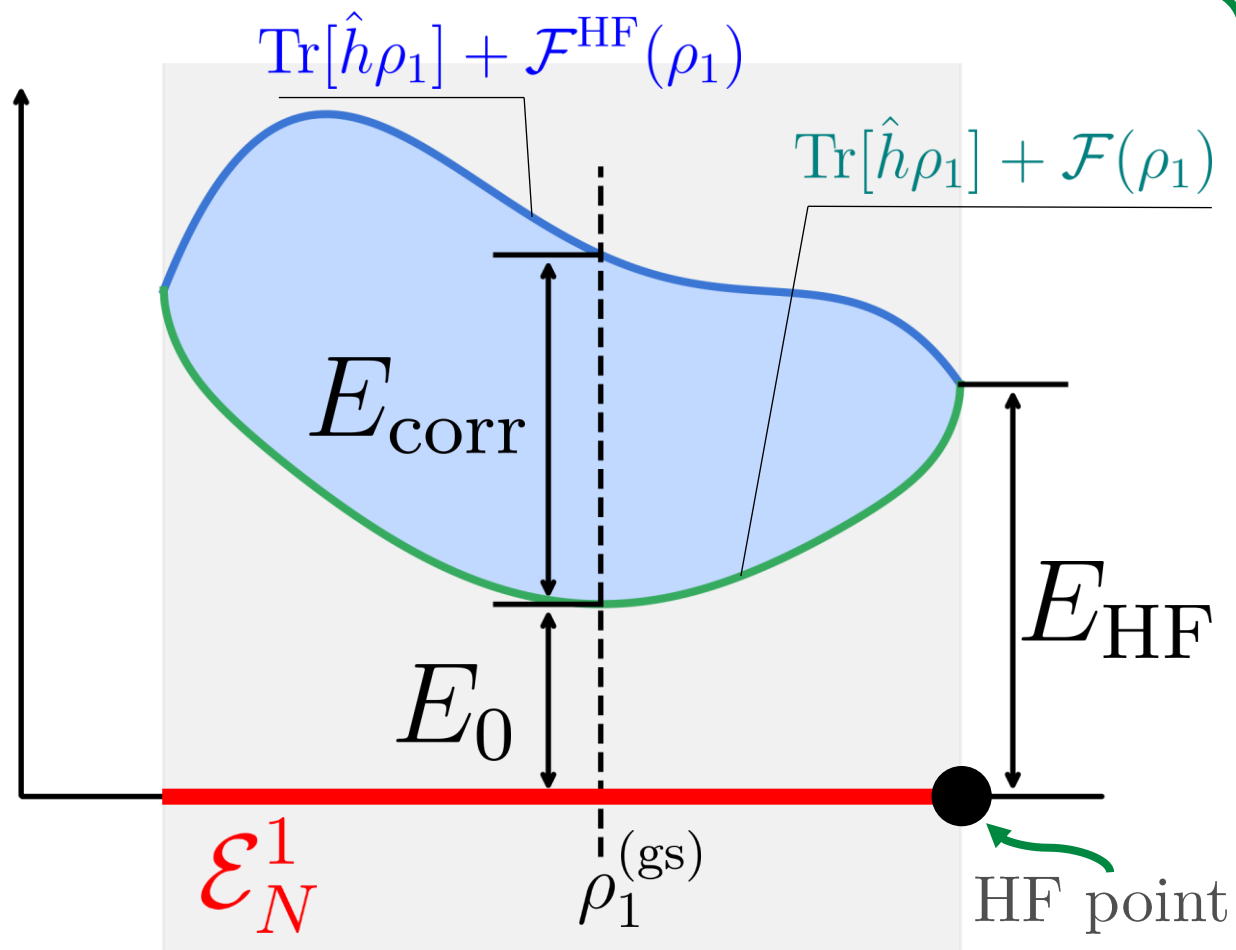
Central challenge of 1RDMFT: obtaining universal functional $\mathcal{F}(\rho_1)$

$$\mathcal{F}(\rho_1) = \mathcal{F}^{\text{HF}}(\rho_1) + \mathcal{F}^{\text{c}}(\rho_1)$$

Key properties of $\mathcal{F}^{\text{c}}$:

- Non-negativity (of $-\mathcal{F}^{\text{c}}$)
- Vanishes for uncorrelated states
- Encode correction beyond HF

→ $\mathcal{F}^{\text{c}}$ behaves like **correlation measure** in QIT sense!



Roadmap

Goal: Accurate description of **strongly correlated** molecular systems

Approach:

Correlation functional is measure of correlation – quantified as energy



QIT provides nicely behaving correlation measures $\eta(\rho_1)$



Approximate $\mathcal{F}^c$ as simple function of η



Definition of *correlation measure functional theory*

(at HF cost level)

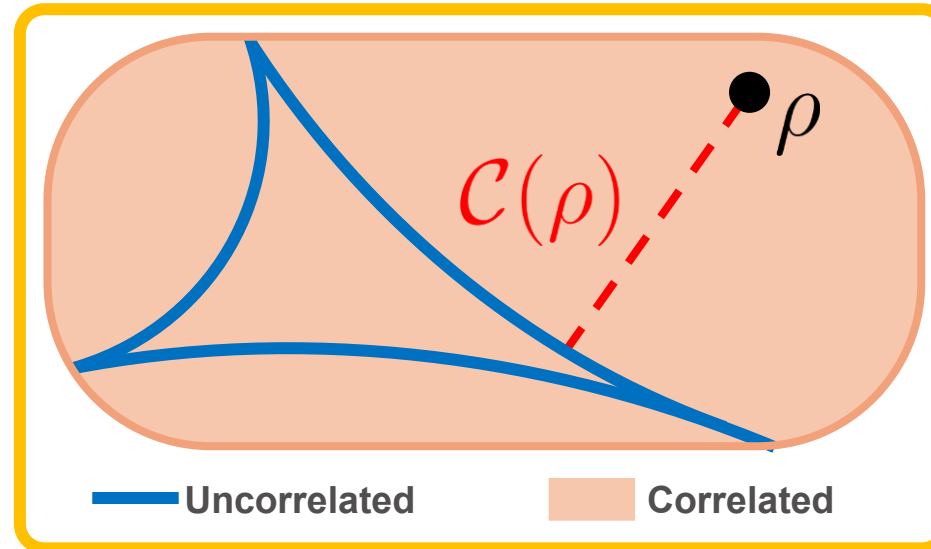


QIT provides correlation measure:

Non-Freeness

Non-freeness: Quantifies structure of some state ρ beyond the uncorrelated state ω_ρ with the same 1-RDM.

Geometric picture behind non-freeness [5-9]



Minimal *distance* from the set of free states

$$\mathcal{C}(\rho) = \eta(\rho_1) = -\text{Tr}_1 [\rho_1 \ln(\rho_1) + (1 - \rho_1) \ln(1 - \rho_1)]$$

[5] – A.D. Gottlieb, N.J. Mauser, *Phys. Rev. Lett.* **95**, 123003 (2005).

[6] – A.D. Gottlieb, N.J. Mauser, *Int. J. Quantum Inf.* **5**, 815 (2007).

[7] – C.J. Turner, K. Meichanetzidis, Z. Papic J.K. Pachos, *Nat Commun* **8**, 14926 (2017).

[8] – L. Ding and C. Schilling, *J. Chem. Theory Comput.* **16**, 4159 (2020)

[9] – D. Aliverti-Piuri, K. Chatterjee, L. Ding, K. Liao, J. Liebert, C. Schilling, *Faraday Discuss.* **254**, 76-106 (2024).

Correlation measure as the fundamental variable [10]

Correlation measure η

$$\rho_N \longmapsto \rho_1 \longmapsto \eta \in \mathbb{R}$$

Free energy functional

$$G_0(\hat{h}, \kappa) = \min_{\eta} \left\{ \min_{\rho_1 \mapsto \eta} \left[\text{Tr}_1[\hat{h}\rho_1] + \mathcal{F}(\rho_1) \right] - \kappa \eta \right\}$$


$$G_0(\hat{h}, \kappa = 0) = E_0(\hat{h})$$

with an internal part

$$\mathcal{G}_{\hat{h}}(\eta) \equiv \min_{\rho_1 \mapsto \eta} \left\{ \text{Tr}_1[\hat{h}\rho_1] + \mathcal{F}(\rho_1) \right\}$$

Universal functional
from 1RDMFT

Correlation measure as the fundamental variable [10]

Correlation measure η

$$\rho_N \longmapsto \rho_1 \longmapsto \eta \in \mathbb{R}$$

Free energy functional

$$G_0(\hat{h}, \kappa) = \min_{\eta} \left\{ \min_{\rho_1 \mapsto \eta} \left[\text{Tr}_1[\hat{h}\rho_1] + \mathcal{F}(\rho_1) \right] - \kappa \eta \right\}$$

$$\mathcal{G}_{\hat{h}}(\eta) \equiv \min_{\rho_1 \mapsto \eta} \left\{ \text{Tr}_1[\hat{h}\rho_1] + \mathcal{F}(\rho_1) \right\}$$

Kohn-Sham-*like* reference system

$$\mathcal{G}_{\hat{h}}^{\text{HF}}(\eta) \equiv \min_{\rho_1 \mapsto \eta} \left\{ \text{Tr}_1[\hat{h}\rho_1] + \mathcal{F}^{\text{HF}}(\rho_1) \right\}$$



Implies correlation functional:

$$\mathcal{G}_{\hat{h}}^{\text{c}}(\eta) = \mathcal{G}_{\hat{h}}(\eta) - \mathcal{G}_{\hat{h}}^{\text{HF}}(\eta)$$

Approximations to the correlation functional

$$\mathcal{G}_{\hat{h}}^c(\eta) = \mathcal{G}_{\hat{h}}(\eta) - \mathcal{G}_{\hat{h}}^{\text{HF}}(\eta)$$

$$\kappa_{\hat{h}} = \frac{\partial \mathcal{G}_{\hat{h}}(\eta)}{\partial \eta} = \boxed{\frac{\partial \mathcal{G}_{\hat{h}}^{\text{HF}}(\eta)}{\partial \eta}} + \boxed{\frac{\partial \mathcal{G}_{\hat{h}}^c(\eta)}{\partial \eta}}$$

$\kappa_{\hat{h}}^{\text{HF}}$ $\kappa_{\hat{h}}^c$

1st order approximation:

$$\mathcal{G}^c(\eta) = -\kappa^c \eta - b^c$$

$$\kappa \longrightarrow 0$$

$$E_0 = \min_{\rho_1} \left\{ \text{Tr}_1 [\hat{h} \rho_1] + \mathcal{F}^{\text{HF}}(\rho_1) - \kappa^c \eta(\rho_1) - b^c \right\}$$

How can first order approximations be found?

Trade-off:

complexity of
the functional

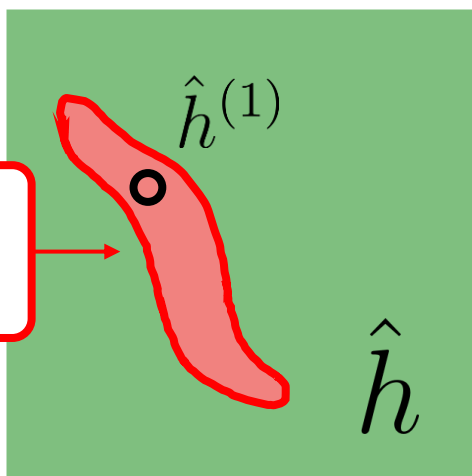


complexity of the
variational domain

$$\mathcal{G}^c = -\kappa^c \eta - b^c$$

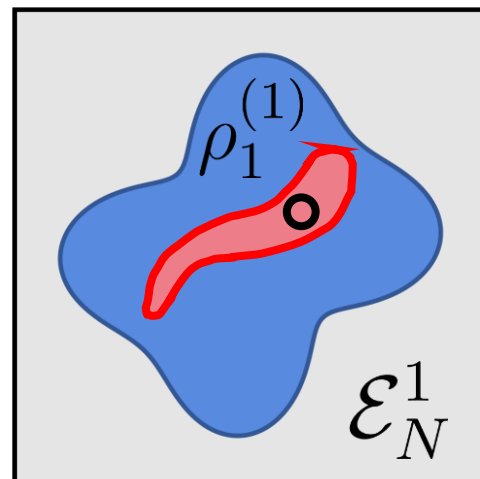
Universality
implies scope

space of $\hat{h}$



space of

$\hat{h}$ -representable ρ_1



$\mathcal{G}^c$

(κ, b)

$(\eta^{(1)}, \mathcal{G}^{c(1)})$

η

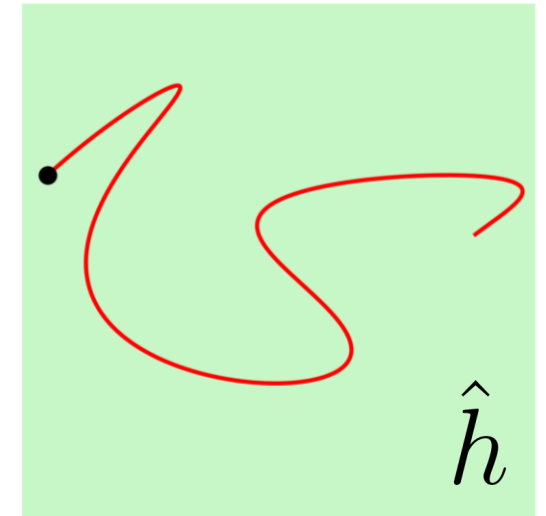
In general, variational domains are complicated!



Dissociation of molecules:



$R = 0.0$ Bohr

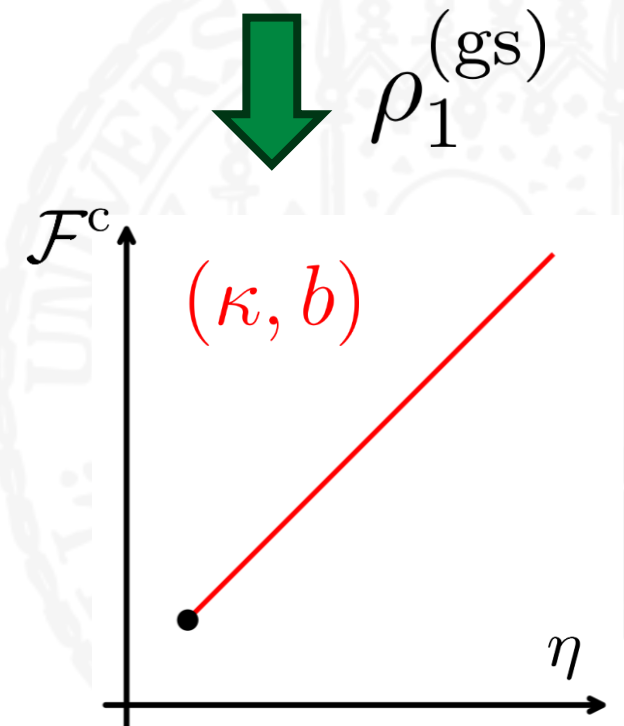


For ground states ρ_1 [11]

$$\mathcal{F}^c(\rho_1) \approx -\kappa^c \eta(\rho_1) - b^c$$

QIT: non-freeness as correlation measure in **QChem**

$$\eta(\rho_1) = -\text{Tr}_1 [\rho_1 \ln(\rho_1) + (1 - \rho_1) \ln(1 - \rho_1)]$$

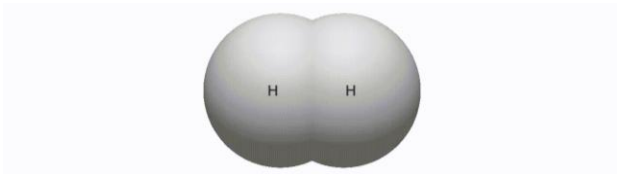
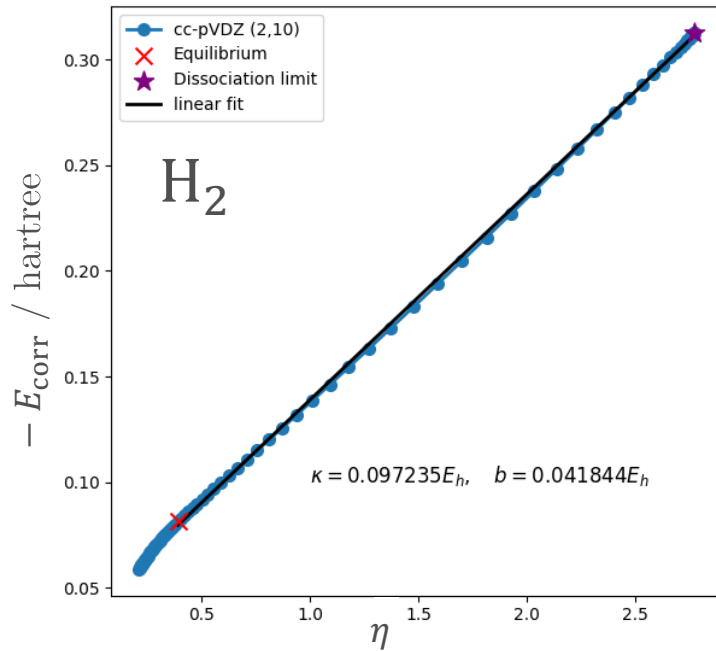


Critical assessment of *Correlation conjecture* [12]

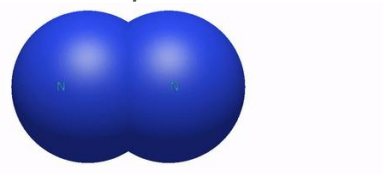
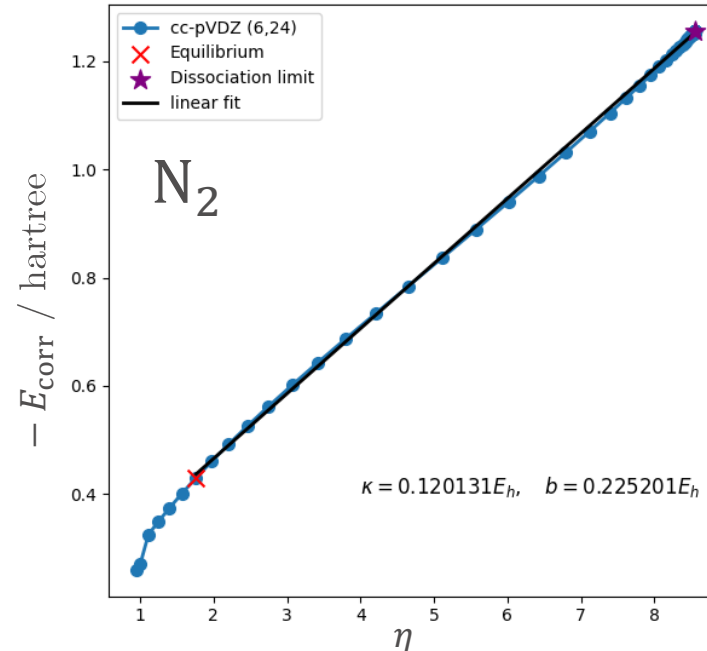
Homolytic dissociation processes

$$\mathcal{F}^c(\rho_1) \approx -\kappa^c \eta(\rho_1) - b^c$$

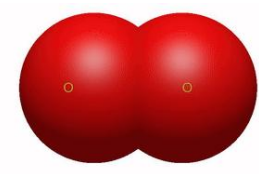
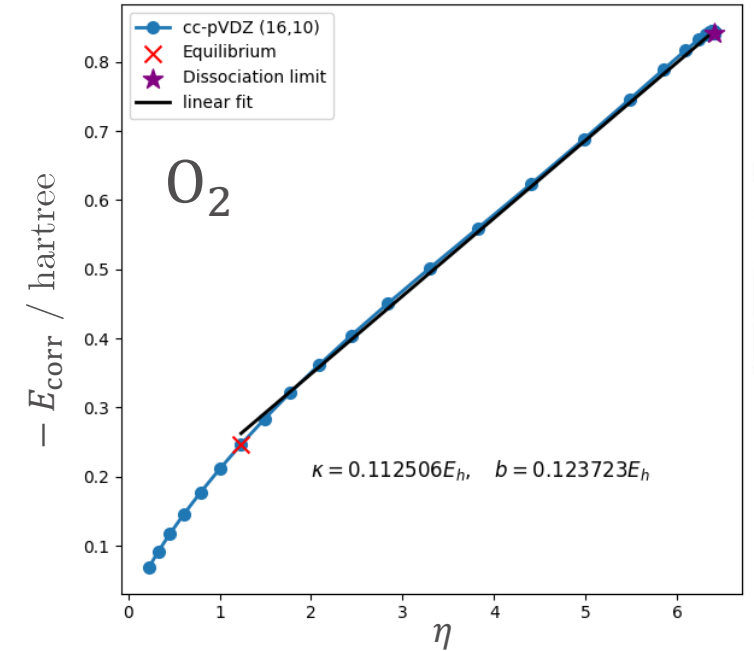
E_{corr} vs. η for ground state of H_2 at different inter-nuclear distances.



E_{corr} vs. η for ground state of N_2 at different inter-nuclear distances.

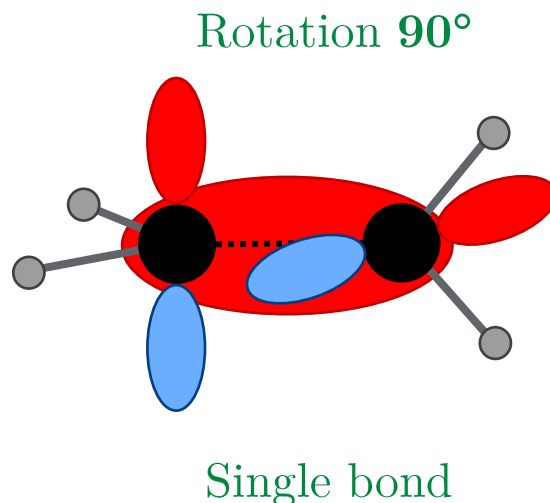
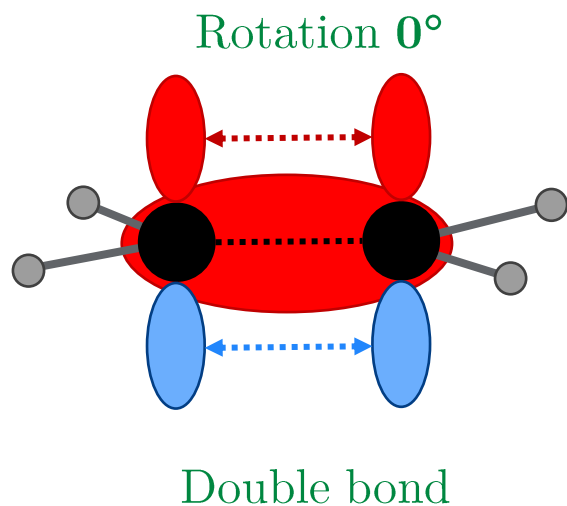
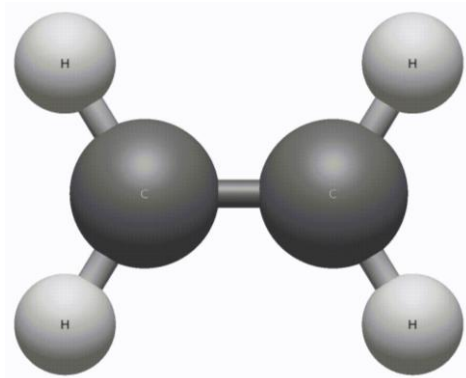
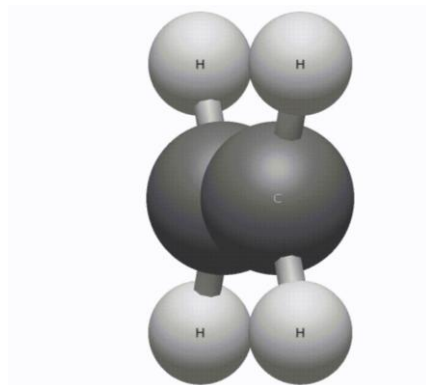


E_{corr} vs. η for ground state of O_2 at different inter-nuclear distances.

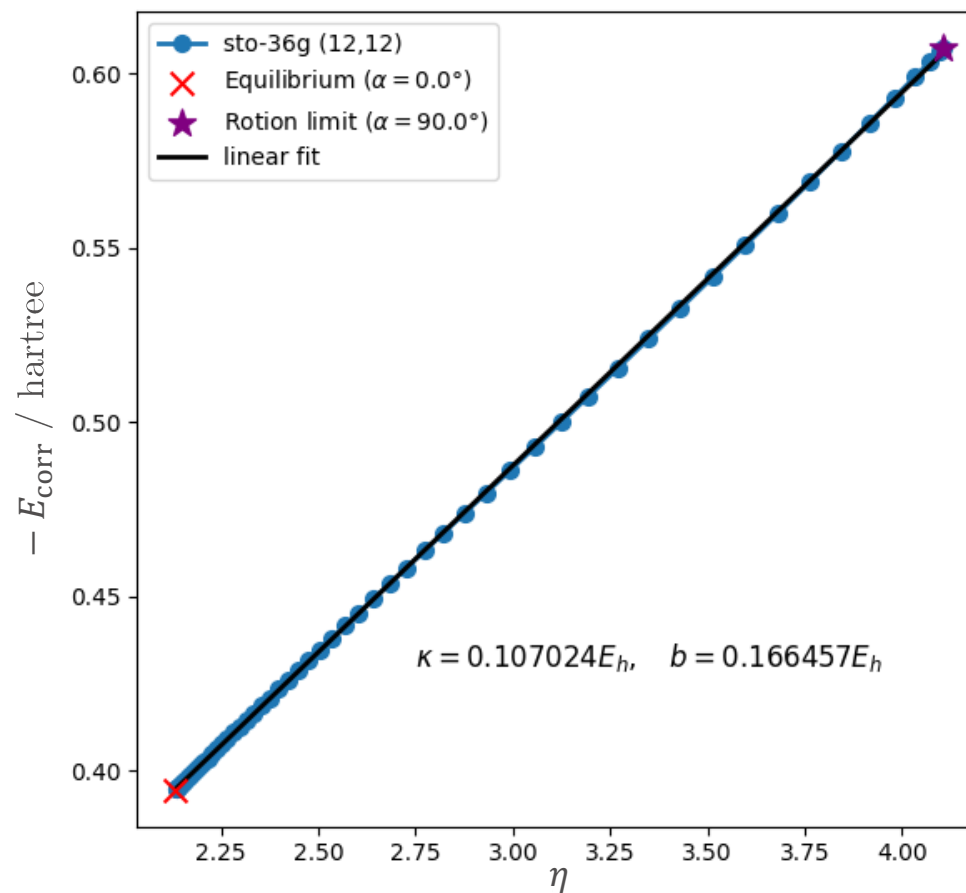


Ethylene C_2H_4

$$\mathcal{F}^c(\rho_1) \approx -\kappa^c \eta(\rho_1) - b^c$$



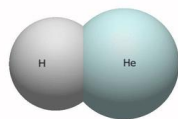
E_{corr} vs. η for ground state of C_2H_4 at different rotation angles α .



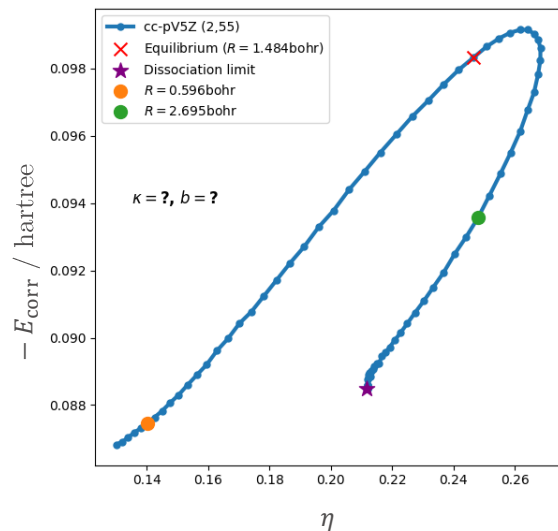
Negative results

$$\mathcal{F}^c(\rho_1) \approx -\kappa^c \eta(\rho_1) - b^c$$

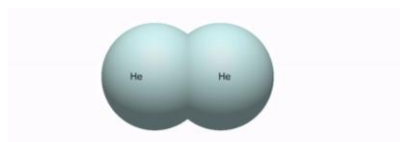
HeH⁺



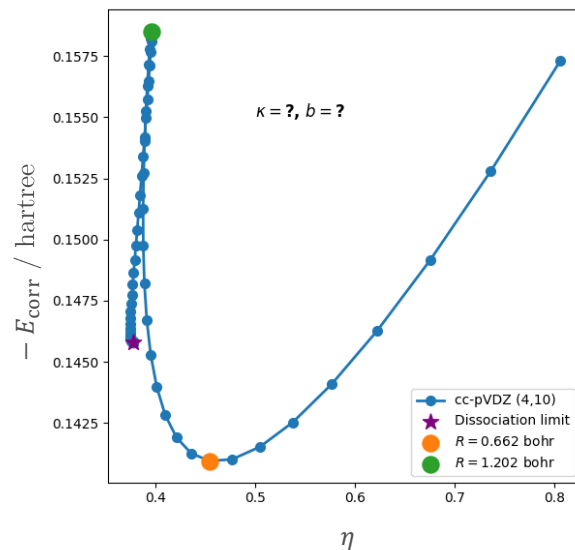
E_{corr} vs. η for ground state of HeH⁺ at different inter-nuclear distances.



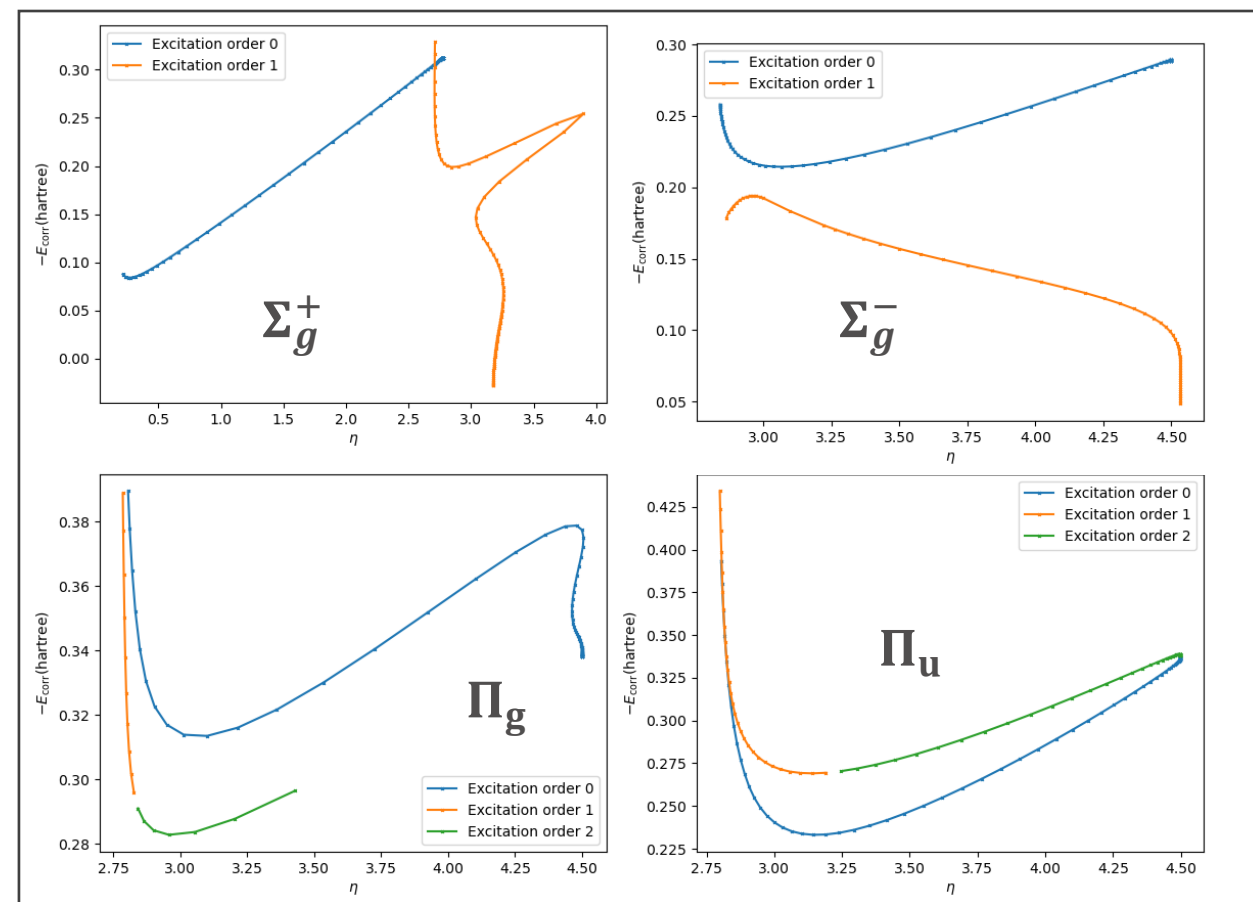
He₂



E_{corr} vs. η for ground state of He₂ at different inter-nuclear distances.



Excited states in H₂



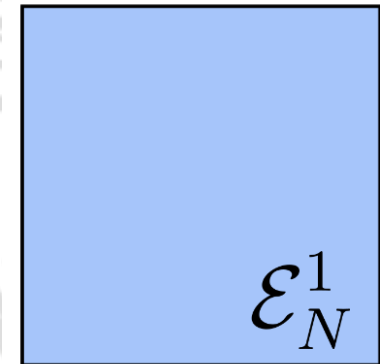
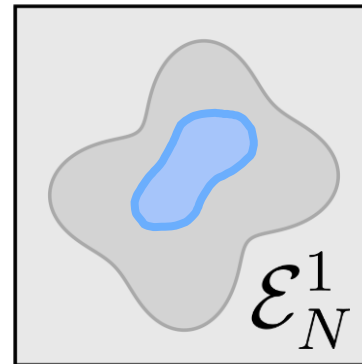
Application of the correlation conjecture in *iDMFT* [13]

Non-freeness: $\eta(\rho_1) = -\text{Tr}_1 [\rho_1 \ln(\rho_1) + (1 - \rho_1) \ln(1 - \rho_1)]$

iDMFT 1RDM functional:

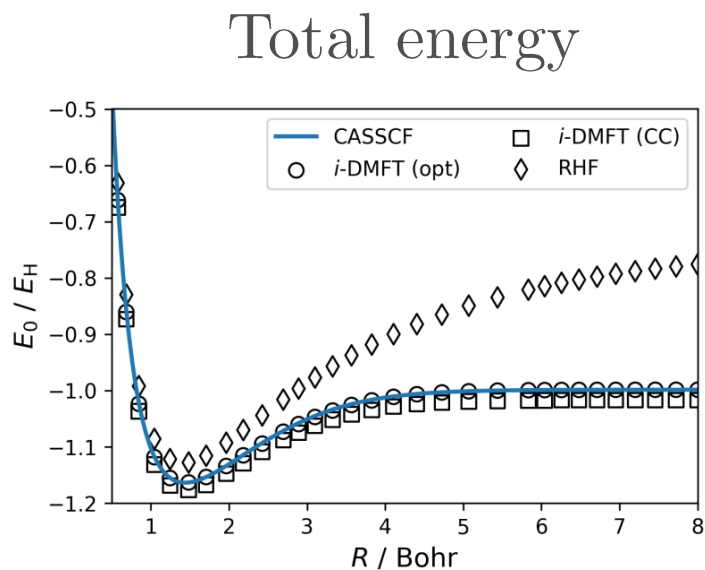
$$E_0 = \min_{\rho_1} \left\{ \text{Tr}_1 [\hat{h} \rho_1] + \mathcal{F}^{\text{HF}}(\rho_1) - \kappa^c \eta(\rho_1) - b^c \right\}$$

Issue of the domain:

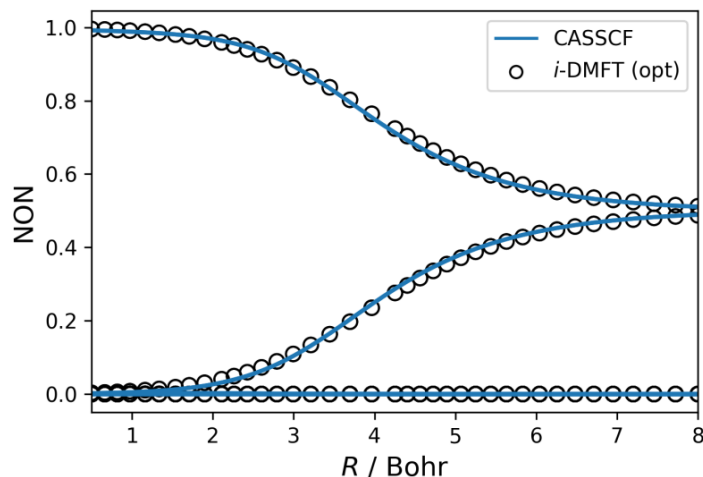


Critical assessment of *iDMFT*

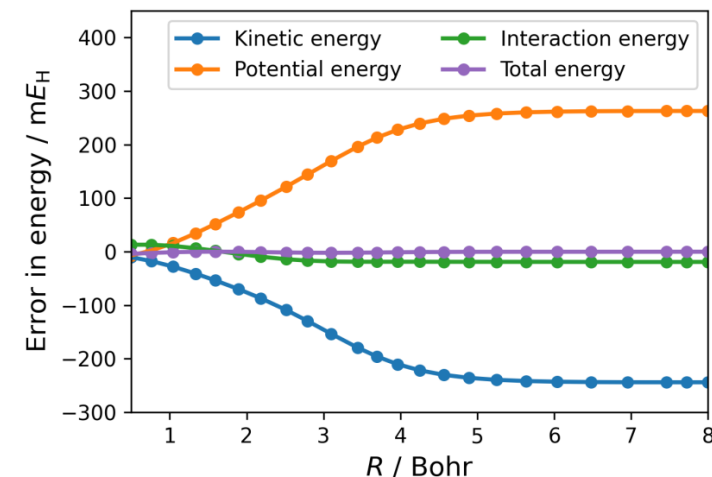
H₂



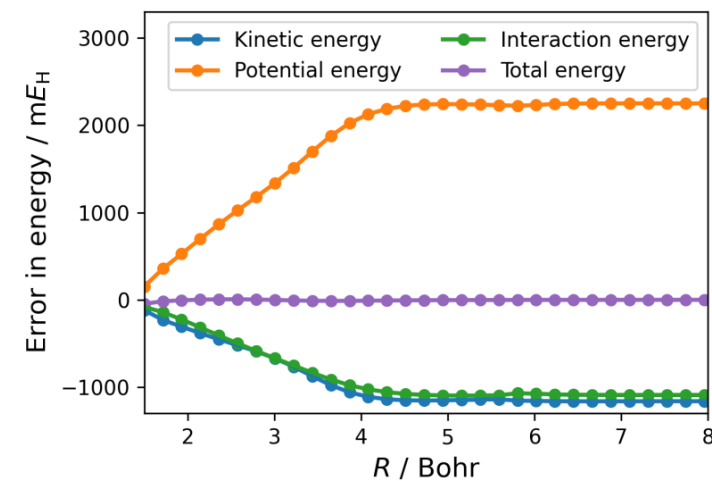
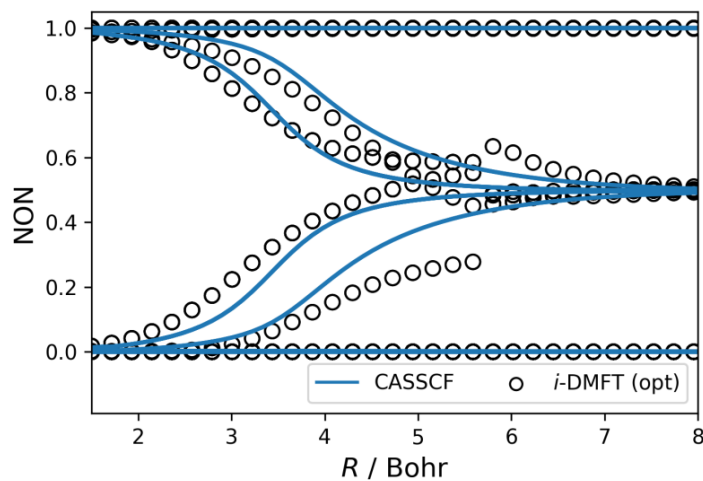
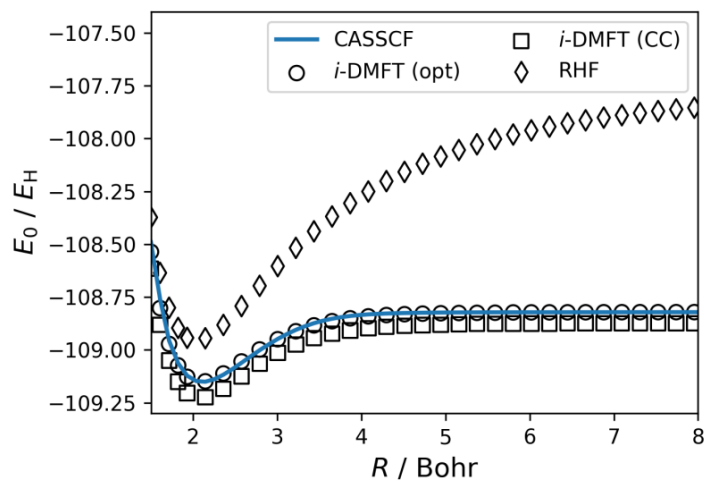
Occupation numbers



Individual energy terms



N₂



Summary

Correlation measure functional theory:

- **no structured approach for construction** of functionals unlike *scope*.
- Requires QIT-influx for sensible correlation measures

iDMFT implementation:

- Captures **strong correlation** at cost of Hartree-Fock
- Where *correlation conjecture* valid $\rightarrow$ **iDMFT recovers energy**
- Other quantities (1RDM, individual energy terms) not approximated well

Outlook:

- Replace HF starting point by functional based on H_2 dissociation reference

Thank you for your attention!

Schilling group



Postdoc/PhD positions

e.g., *BeyondDFT*





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Repository of additional slides.

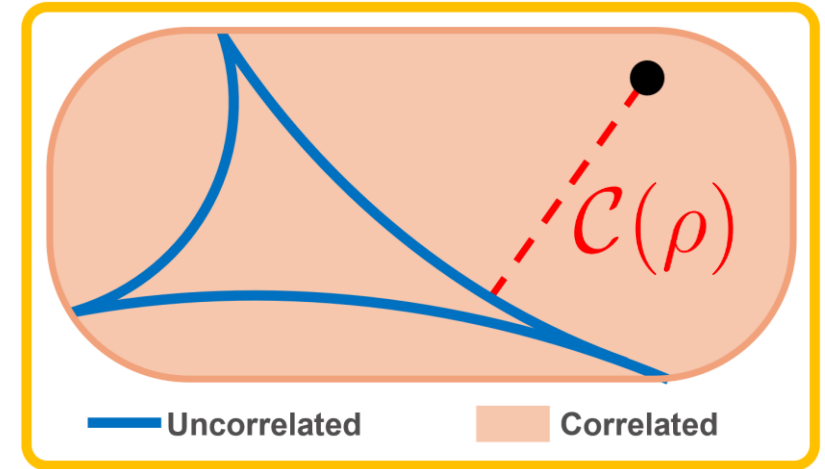




QIT: measure of non-freeness

$$\eta(\rho_1) = -\text{Tr}_1 [\rho_1 \ln(\rho_1) + (1 - \rho_1) \ln(1 - \rho_1)]$$

Minimal *distance* from the set of free states measured in terms of *quantum relative entropy*.



Quantum relative entropy between states σ and ρ :

$$D(\rho||\sigma) = \text{Tr}[\rho \log(\rho)] - \text{Tr}[\rho \log(\sigma)]$$

where $D(\rho||\sigma) = 0 \Leftrightarrow \rho = \sigma$

Interpretation:

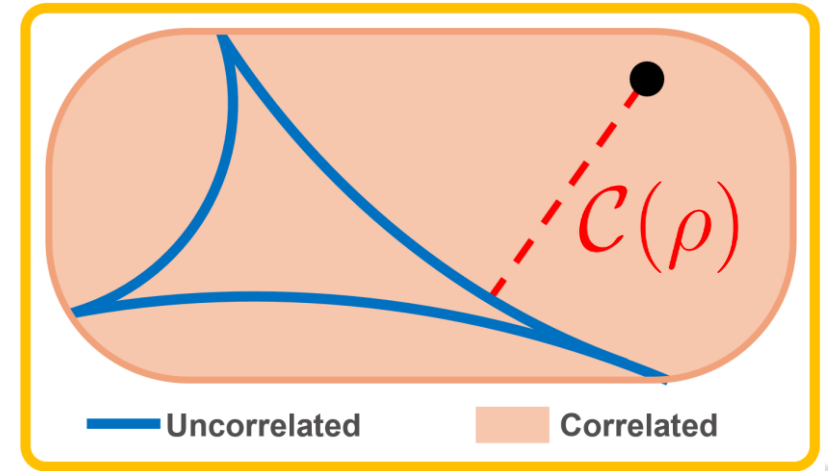
- Quantifies difficulty to distinguish state ρ from reference state σ .
- n copies: error rate of falsely assuming σ , while infact ρ : $\exp[-nD(\rho||\sigma)]$.



QIT: measure of non-freeness

$$\eta(\rho_1) = -\text{Tr}_1 [\rho_1 \ln(\rho_1) + (1 - \rho_1) \ln(1 - \rho_1)]$$

Minimal *distance* from the set of free states measured in terms of *quantum relative entropy*.



Uncorrelated (free) states:

$$\omega \propto \exp \left[\alpha \sum_{i,j} h_{ij} \hat{a}_i^\dagger \hat{a}_j \right]$$

Non-freeness: How much extra structure does ρ contain beyond the best possible free-fermion description compatible with its 1-RDM?

$$\mathcal{C}(\rho) = \min_{\omega \in \mathfrak{F}} D(\rho || \omega) \left(= \eta(\rho_1(\rho)) \right)$$

$$\mathcal{F}^c(\rho_1) \approx -\kappa^c \eta(\rho_1) - b^c$$

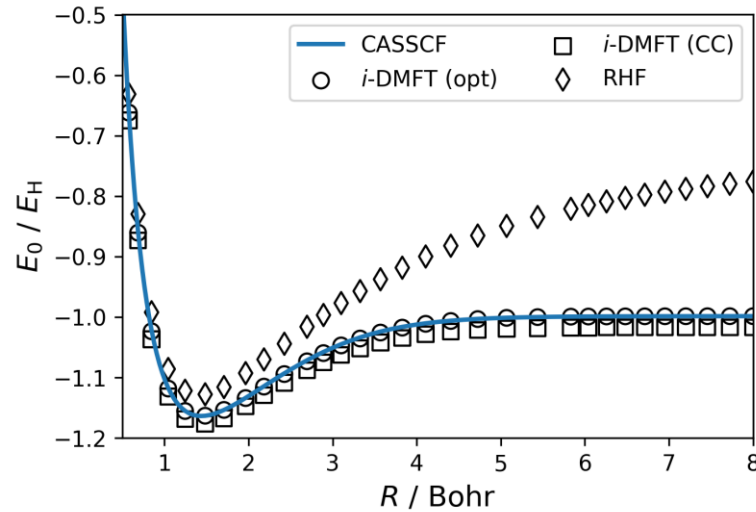
3 criteria from investigations
of correlation conjecture:

1. The electron reorganization is merely a redistribution within pairs of orbitals, e.g., bonding/antibonding orbitals.
2. The bond breaking is not completely heterolytic.
3. The electronic states involved are electronic ground states and there are no avoided crossings.

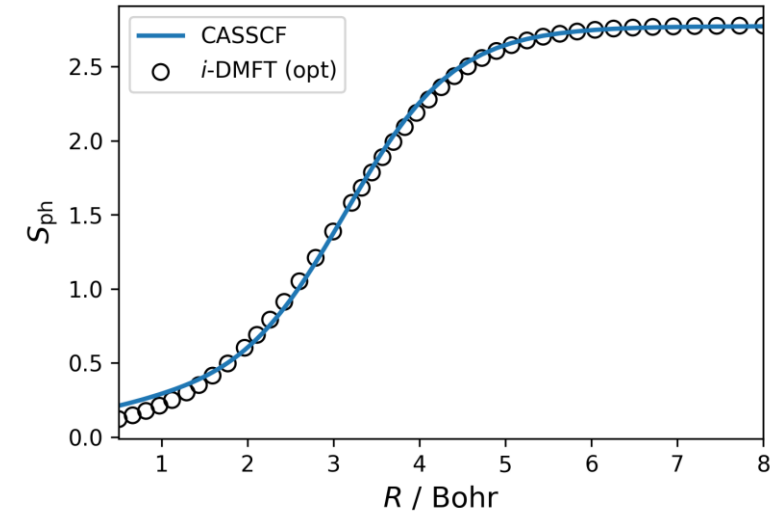
Critical assessment of *iDMFT*

H₂

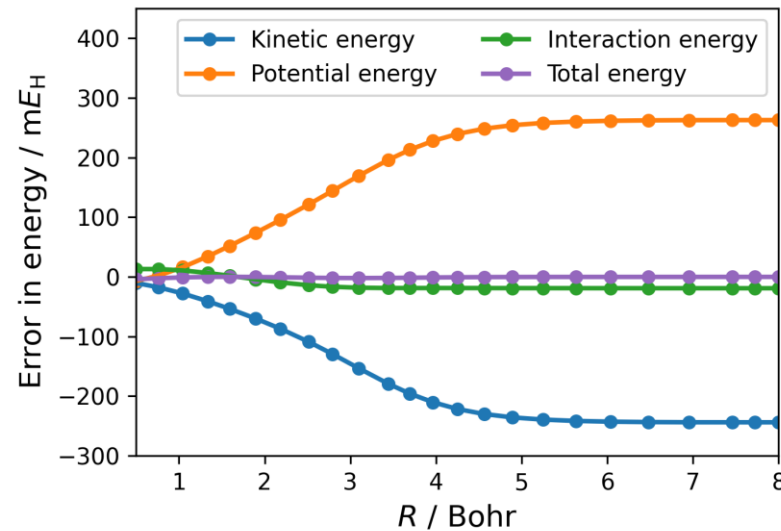
Total energy



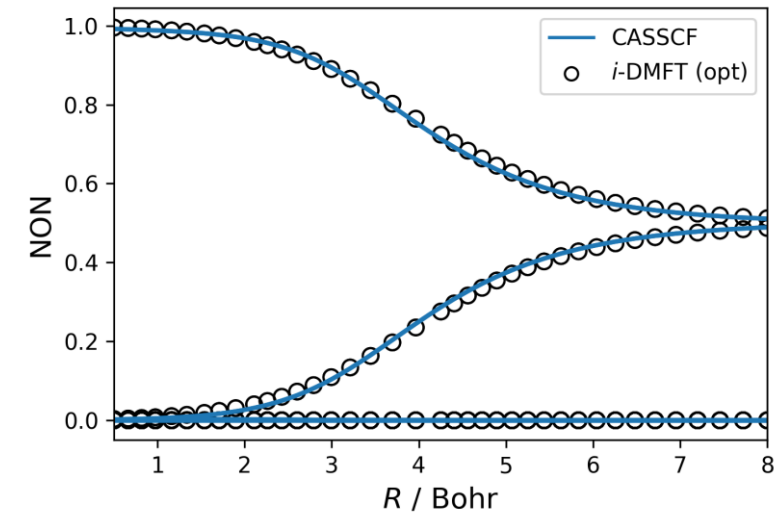
Non-freeness



Individual energy terms



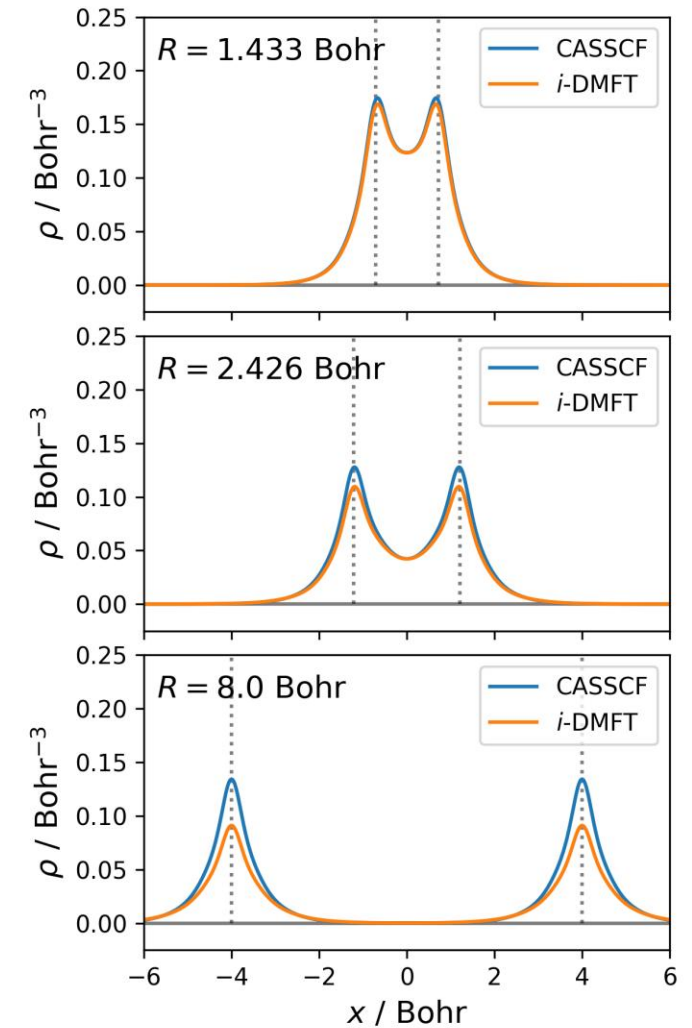
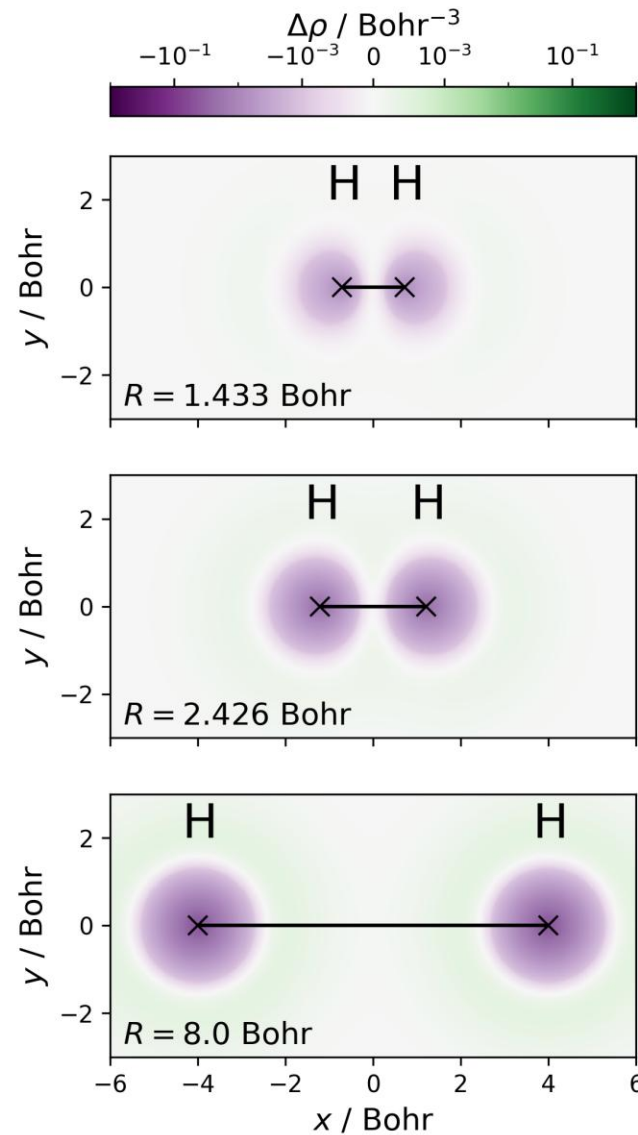
Natural occupation numbers



Critical assessment of *iDMFT*

H₂

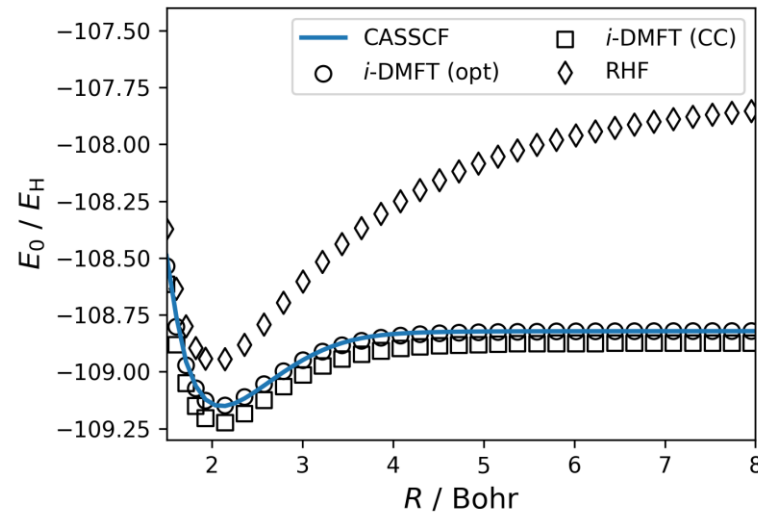
Error in electronic density



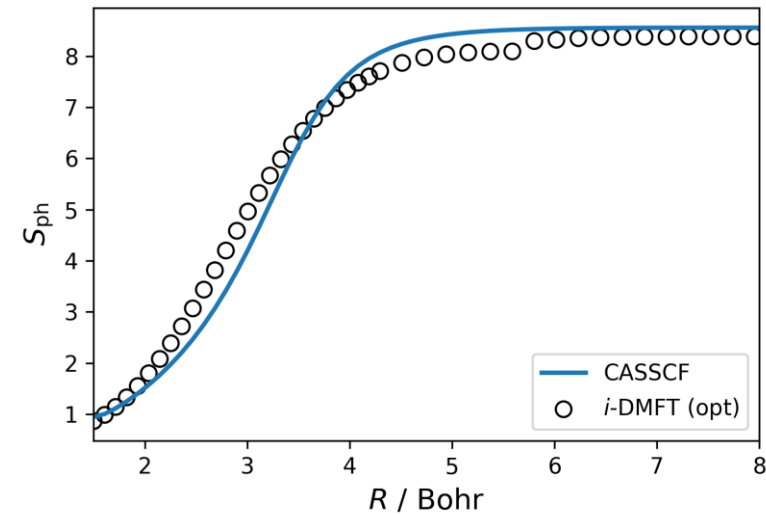
Critical assessment of *iDMFT*

N₂

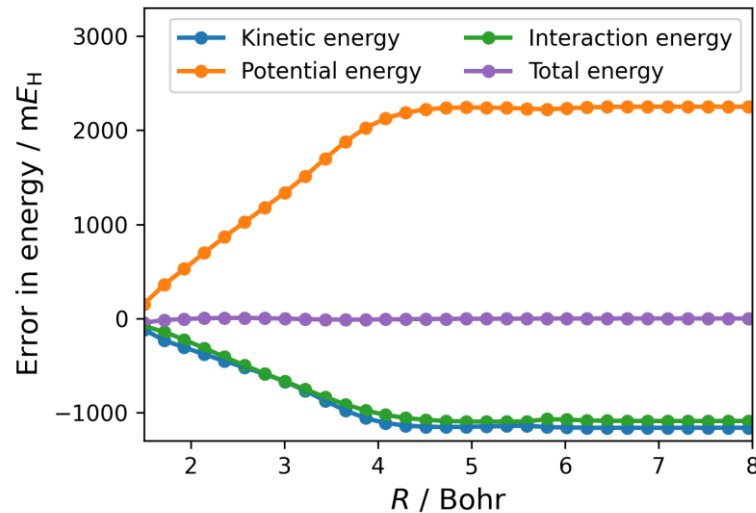
Total energy



Non-freeness



Individual energy terms



Natural occupation numbers

